

Cross-task consistent multi-task learning for fine-grained crop leaf disease diagnosis and severity grading

Jin Li^{1st, *}

Dundee International Institute of Central
South University
Central South University
Changsha, China

* Corresponding author:
7803230111@csu.edu.cn

Yuqiu Zhang^{1st}

Dundee International Institute of Central
South University
Central South University
Changsha, China

7803230104@csu.edu.cn

Shucen Guo

Dundee International Institute of Central
South University
Central South University
Changsha, China

7803230109@csu.edu.cn

Abstract—In smart agriculture, diagnosing crop leaf diseases involves greater complexity than merely predicting isolated categories: a practical system must also be able to assess whether the infection is of mild or severe severity. However, many current studies treat the optimization of fine-grained recognition and severity assessment as two separate tasks, which is likely to increase implementation costs and lead to potentially contradictory results. This paper presents a cross-task consistent multi-task architecture for diagnosing crop leaf diseases and grading severity. The model adopts a ConvNeXt-Tiny backbone, inserts a CBAM attention block to emphasize discriminative lesion regions, and builds parallel heads for 61-class disease recognition and three-level severity grading. Transforming the disease posterior using a consistency regularizer creates an expected severity distribution, which is aligned with the severity branch in training. Experiments on a dataset containing 31,541 training images and 4,533 validation images from 10 crops show that the proposed model reaches 87.60% accuracy and 0.8564 macro F1-score on fine-grained disease classification, while obtaining 92.98% accuracy and 0.9338 macro F1-score on severity grading. The results indicate that precise and consistent predictions for real-world agricultural diagnosis can be delivered using a single, lightweight framework.

Keywords—multi-task learning; cross-task consistency; computer vision in agriculture; fine-grained image classification; attention mechanism

I. INTRODUCTION

Plant diseases remain one of the major sources of crop yield loss, quality degradation, and unnecessary pesticide consumption. However, many agricultural practitioners still rely on manual inspections to identify diseases, which is time-consuming. Thus, as plant disease research has expanded rapidly, image-based diagnosis for smart farming has attracted increasing attention. Recent studies have shown that deep learning has become the dominant approach for automated plant disease analysis [1].

Initial image-based methods primarily dealt with the problem of single-task classification, where a network predicts either the disease type or the general health status of a crop from

a leaf image [2]. While such methods have a powerful performance on controlled data sets, they do not capture the complexity of real-life decision-making. Therefore, recent research has focused on disease grading, segmentation-based severity estimation, and multi-task learning. Examples include apple leaf severity grading with segmentation and watershed refinement [5], multi-task pipelined interpretable analysis of rice bacterial leaf blight [6], and combined grading and classification of diseases in citrus crops [8]. This approach creates additional complexity for inference and may produce semantically incompatible results.

To address this issue, this paper develops a unified conference-style framework for fine-grained crop leaf disease diagnosis and severity grading. The research utilizes a large multi-crop image dataset that contains 61 disease types across 10 crop species. Rather than maintaining the original competition-based fragmentation into several subtasks, the approach is reframed into a single multi-task learning framework where one branch determines the disease class, and the other branch estimates the severity. The two branches share a ConvNeXt-Tiny feature extractor [3], while a CBAM attention module [4] strengthens lesion-focused representation learning.

The main contributions of this paper are threefold. First, a compact shared-backbone architecture is introduced that concurrently addresses 61-class disease recognition and three-tier severity classification in a single model. Also, a cross-task consistency loss is proposed that modifies disease probabilities to an expected severity distribution and aligns it to the severity branch because of loss optimization. Finally, experimental results show that the proposed method improves fine-grained disease recognition and maintains strong severity grading performance, making it suitable for on-field disease classification.

A. Related Work and Gap Analysis

End-to-end deep networks are the latest advancement in the typical pipeline of developing tailored descriptors for specific applications. Performance differences among classical convolutional neural networks, which stem from their varying

ability to learn lesion textures, shapes, and color features from leaf images, were most prominent in early studies. Nevertheless, due to the varying visual symptoms and the shift in the growth stage, the lighting conditions, and the angle of capture, the phenomenon of visual crop disease recognition remains a major challenge.

A second line of research focuses on severity assessment rather than disease classification. Some approaches determine severity via segmentation of lesions or measurement of diseased areas, which provides strong interpretability but typically demands pixel-level annotations and imaging conditions that are tightly controlled. Other approaches view severity in terms of classification, with typical classes of mild, moderate, and severe. While this construction is easier to train in large numbers, it still demands that the model learn crop-agnostic symptom progression, rather than simply memorizing the crop identity.

A third relevant direction is few-shot and low-data learning. Recent studies on foundation models and metric learning show that plant disease recognition can leverage transfer learning, prototype construction, and cross-domain representation reuse [9]. These approaches capture useful characteristics of agricultural datasets due to widespread class imbalance and long-tail categories. However, an accurate disease classifier is insufficient if it does not also provide insights into how urgent and advanced the disease is.

Multi-task learning is, therefore, attractive, but current studies still have two shortcomings. First, most frameworks utilize a shared backbone, but they do not explicitly limit the outputs of the pertinent tasks. Second, some studies discuss disease and severity predictions separately, even though disease labels entail associated severity classifications. This paper attempts to bridge this gap by implementing a lightweight consistency mechanism between the disease and severity branches, while ensuring that the model remains sufficiently compact to be used for practical agricultural diagnosis.

II. DATASET AND TASK DEFINITION

The crop leaf dataset has 31,541 images for training and 4,533 images for validation, spread across 10 crop species and 61 fine-grained disease classifications that include disease-free leaves, mild disease manifestations, and leaves with severe disease manifestations. This makes the task particularly challenging for two primary reasons: the disease symptom intensity and the presence of visually related categories across different crop species.

The dataset covers 27 types of diseases in 10 crops, meaning that the same disease family might manifest through different textures and color distributions across different types of plants. Conversely, different diseases can yield visually identical lesions, making direct image memorization infeasible and forcing the model to learn implicit, non-obvious shared feature representations.

In constructing a severity task, the 61 disease categories were restructured into three severity levels: Healthy, General, and Serious. Categories marked as healthy were classified into the Healthy class. Disease categories that had mild or medium visual symptoms were placed into General, while categories that describe advanced progression or descriptions that are clearly

severe were placed into Serious. The remapped severity space now has 10 healthy categories, 25 general categories, and 26 serious categories.

This remapping is useful in practical applications, as it represents real field choices. A healthy prediction means no action is required. A general prediction means watchful waiting or limited intervention. A serious prediction means an immediate intervention is warranted. Thus, the multi-task framework integrates both agronomic reasoning and subsequent recommendation logic, avoiding the need for an extra post-processing model.

Before training, all images were subjected to basic quality control, including duplicate screening, blur checking, exposure review, and geometric normalization. Images were resized to 224×224 , normalized with ImageNet statistics, and augmented using horizontal flips, random rotations, mild color perturbation, and label-preserving mixing strategies. For severity grading, color augmentation was kept conservative so that lesion color and spread remained informative. Key dataset statistics are summarized in Table 1.

TABLE I. DATASET STATISTICS AND TASK SETTINGS

Item	Value
crop species	10
disease categories	61
training images	31,541
validation images	4,533
severity mapping	Healthy (10), General (25), Serious (26)
input resolution	224×224

Let I denote an input leaf image, y_d denote the disease label, and y_s denote the severity label. The goal of the proposed model is to learn a mapping $f(I) \rightarrow (y_d, y_s)$, where y_d belongs to one of the 61 fine-grained classes and y_s belongs to {Healthy, General, Serious}

A. Preprocessing and Label Construction

Before model training, explicit quality-control heuristics were applied to the full dataset. Blur was estimated from Laplacian variance so that strongly defocused samples could be reviewed, while mean intensity statistics were used to identify clearly overexposed and underexposed images. These steps are simple, but they matter because agricultural datasets often contain repeated handheld captures under unstable illumination.

Duplicate detection is equally important. Near-identical images can make validation performance look artificially strong if they leak into both splits. To reduce this risk, perceptual similarity screening was applied during data preparation. After quality filtering, all images were resized to 224×224 and normalized with ImageNet statistics, which keeps the pipeline compatible with standard pretrained ConvNeXt models.

The severity remapping also had to address some specific labels that do not explicitly indicate severity. From the original task analysis, Apple Frogeye Spot was treated as a General case because it appears as localized lesions and not as severe systemic destruction. In contrast, the Maize Dwarf Mosaic Virus and the Tomato Mosaic Virus were categorized as Serious because the

visual impact and systemic spread of these two are usually much more pronounced. This kind of rule-based case normalization is very helpful in retaining consistency in labelling across different crops.

The rule-based mapping is simple, but it preserves the semantic link between disease names and severity levels. Compared with automated clustering, it avoids extra hyperparameter tuning and keeps the severity definition aligned with agronomic knowledge, which is more suitable for this task with a clear label structure.

For data augmentation, a combination of horizontal flips, random rotations, resized crops, and conservative color perturbation was applied. In a disease classification task, moderate variation in appearance is beneficial for generalizing the model to account for viewpoint and illumination changes.

With severity grading, however, the photometric augmentation must be more constrained because lesion color and the extent of chlorosis and necrosis are all salient clues. This disparity justified the employment of severity-preserving augmentation as opposed to bold cross-class mixing.

III. PROPOSED METHOD

The overall framework is illustrated in Figure 1. A shared encoder first transforms the input leaf image into a compact representation, after which two task-specific heads predict disease identity and severity level. During training, the two predictions are additionally constrained by a cross-task consistency term. This design keeps the model compact while preserving semantic coherence between outputs.

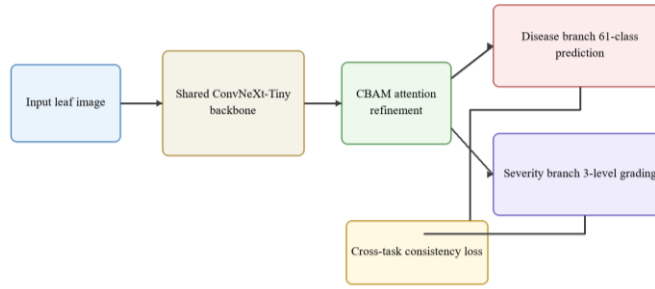


Fig. 1. Overview of the proposed cross-task consistent multi-task framework.

A. Shared Backbone and Attention Refinement

ConvNeXt-Tiny was selected as the shared encoder because it offers a strong balance between representation quality and computational cost [3]. Although Vision Transformer variants are strong in global modeling, this task relies more on stable extraction of lesion texture, boundary shape, and local color variation. A convolutional inductive bias is therefore more suitable here. ConvNeXt-Tiny offers strong representation power while keeping training stable and deployment cost moderate, making it a better shared backbone for this setting. For the shared encoder, we prioritize a balance among hierarchical representation capacity, training stability, and deployment cost. Although recent lightweight backbones such as MobileNetV4 and EfficientViT are attractive from an efficiency perspective, the present task relies more heavily on robust modeling of textures, lesion boundaries, discoloration patterns, and spatial disease spread. ConvNeXt-Tiny inherits the convolutional inductive bias while incorporating modern design principles from recent vision architectures. It provides stronger hierarchical feature extraction with a moderate parameter budget and remains highly compatible with standard pretraining and optimization pipelines. Therefore, ConvNeXt-Tiny is a more suitable shared backbone for our setting, as it better balances fine-grained recognition performance and practical deployability in agricultural scenarios. Compared with heavier transformer-style architectures, ConvNeXt-Tiny is easier to train and deploy while still providing robust hierarchical feature extraction. A comprehensive pre-evaluation of the state-of-the-art lightweight Vision Transformer baseline (DeiT-Tiny) was performed under

identical training pipelines, dataset constraints, and computational budgets. Pre-evaluation results show that DeiT-Tiny achieves comparable performance on the severity grading task (92.12% accuracy) but suffers a significant performance drop in 61-class fine-grained disease classification, with a top-1 accuracy of 83.25% and macro F1-score of 0.7916, 3.49 percentage points and 0.0357 lower than the ConvNeXt-Tiny single-task baseline, respectively.

This performance gap originates from the inherent adaptation of convolutional inductive bias to the task characteristics of agricultural disease diagnosis. The translation-equivariance of convolutional operations inherently matches the spatial locality of core diagnostic cues, including subtle lesion textures, color shifts, and boundary morphology, which are easily disturbed by complex field backgrounds and uneven illumination in the global self-attention mechanism of ViT. Meanwhile, the hierarchical multi-scale feature extraction paradigm of ConvNeXt-Tiny is more compatible with the dual-branch multi-task framework, preserving general low-level features while encoding high-level disease semantics, effectively mitigating inter-task gradient conflicts. For edge deployment in smart agriculture, ConvNeXt-Tiny also maintains more stable inference efficiency on low-quality field images with a moderate parameter budget.

The above results and task-specific analysis support the final selection of ConvNeXt-Tiny with convolutional inductive bias as the shared backbone, which achieves a better balance between fine-grained classification performance, multi-task

compatibility, and practical deployability. To improve sensitivity to lesion boundaries, discoloration, and texture change, a CBAM module is inserted after the backbone output. Placing CBAM after the backbone output enables the network to first preserve low-level texture cues and mid-level edge information, then recalibrate high-level semantic features. Repeated integration of CBAM across multiple stages increases computational overhead, while the resulting performance gain is not guaranteed to be stable or significant. End-of-backbone placement is therefore a more compact trade-off. The shared feature extraction process is written as

$$\mathbf{f} = \Phi_{\text{CBAM}}(\Phi_{\text{ConvNeXt}(l)}), \quad (1)$$

where $\mathbf{f}$ is a 768-dimensional embedding vector. Regarding the placement of the attention module, CBAM is inserted after the backbone output instead of integrating it into every ConvNeXt stage. This choice reflects a trade-off between semantic specificity and model complexity. The early and middle stages of ConvNeXt are better suited for preserving generic edge and texture cues, whereas the final backbone output already contains more complete disease semantics. Repeated insertion of CBAM at multiple stages may provide more frequent feature recalibration, but it also introduces additional computational overhead and may interfere with the backbone’s progressive multi-scale encoding. By contrast, applying CBAM once to the final high-level features allows the model to highlight lesion regions, discoloration boundaries, and sparse symptom areas more directly, while keeping the shared encoder compact and deployment-friendly. Since CBAM itself is a lightweight and general-purpose module, this end-of-backbone design better matches our goal of balancing accuracy and efficiency. In practice, this combination allows the network to use both global crop context and local symptom evidence. The attention module is especially useful when disease classes differ by subtle color shift or sparse lesion distribution.

Another practical reason for this design is parameter efficiency. Instead of building separate encoders for disease naming and severity grading, both tasks reuse the same feature extractor. This lowers memory demand, reduces duplicated computation, and encourages both tasks to learn from the same lesion-centered cues. The resulting model remains compact enough for deployment-oriented scenarios while still preserving strong representational capacity.

B. Disease Classification Branch

The disease branch is designed for 61-class fine-grained recognition. A progressive classifier is used to reduce feature dimension while preserving discrimination capability:

$$p_d = \text{Softmax}(W_{d2}\sigma(W_{d1}\mathbf{f} + b_{d1}) + b_{d2}), \quad (2)$$

where σ denotes nonlinear activation and p_d is the posterior probability over disease classes. The branch uses focal loss with label smoothing to address class imbalance and difficult examples. This choice is appropriate for crop disease datasets because some categories contain sparse lesion patterns and are visually close to neighboring categories.

In fine-grained agricultural recognition, top-1 accuracy alone does not fully describe model quality. A useful disease branch should also provide calibrated class probabilities and stable

ranking among visually similar candidates. The hierarchical fully connected structure, therefore, serves two purposes: it compresses the backbone representation and allows the network to separate coarse disease families before resolving subtle sub-category differences.

The disease branch also serves as the semantic anchor for the entire framework, as the predicted disease posterior is later transformed into an expected severity prior; this branch must remain discriminative but not overconfident. Label smoothing helps prevent brittle decision boundaries, while focal weighting emphasizes visually difficult categories that would otherwise be overshadowed by easier healthy, or strongly expressed classes.

C. Severity Grading Branch and Cross-task Consistency

The severity branch predicts one of three levels: Healthy, General, or Serious. Because this task is coarser than disease recognition, the classifier is narrower:

$$p_s = \text{Softmax}(W_{s2}g(W_{s1}\mathbf{f} + b_{s1}) + b_{s2}), \quad (3)$$

where g denotes nonlinear activation and p_s is the severity posterior. In addition to the main three-class head, the original experimental pipeline used a lightweight auxiliary binary head to distinguish healthy from diseased leaves. This auxiliary signal improves feature separation at the boundary between truly healthy samples and early disease symptoms.

The key challenge is that disease and severity are not independent. Each disease class implies a valid severity level, and a prediction pair should remain compatible. Therefore, we first define a deterministic mapping $M(c)$ that maps each disease class c to its corresponding severity level, and then compute the expected severity distribution induced by the disease branch as

$$p_{\hat{s}} = \sum_c p_{d(c)} e_{M(c)}, \quad (4)$$

where $e_{M(c)}$ is the one-hot vector of the severity level associated with class c . A Kullback-Leibler divergence term is then used to align the severity branch to this expected distribution. For the cross-task consistency term, Kullback-Leibler (KL) divergence is adopted instead of mean squared error (MSE) calculated on logits, as the alignment target is fundamentally the relationship between probability distributions, not raw unnormalized logit values. The 61-way disease posterior is first converted into a disease-induced severity prior through a deterministic mapping and then compared with the three-class severity distribution predicted by the severity branch. KL divergence directly measures the deviation between two probability distributions and explicitly encourages semantic consistency during optimization. In contrast, MSE only measures numerical distance and does not capture distributional constraints, confidence structure, or soft-label semantics. Therefore, KL divergence is a more appropriate regularizer for our cross-task consistency objective, as it strengthens semantic alignment between the two branches without sacrificing fine-grained disease discrimination:

$$L_{\text{cons}} = \text{KL}(p_{\hat{s}} \parallel p_s). \quad (5)$$

This regularization does not force the two tasks to collapse into each other. Instead, it allows the disease branch to keep fine-grained discrimination while encouraging the severity branch to

stay consistent with disease semantics. The mechanism is lightweight, differentiable, and easy to add to a shared-backbone architecture.

The benefit of this design is most visible in borderline cases. A leaf may contain a limited lesion area yet still belong to a disease class that is typically serious once it emerges. Likewise, a visually mild sample should not be paired with a disease posterior that strongly supports a class associated with severe symptoms. The cross-task term suppresses these implausible combinations and provides a cleaner diagnostic signal.

D. Training Objective and Implementation Details

The final optimization target combines disease loss, severity loss, and consistency regularization:

$$\mathcal{L} = \mathcal{L}_{\text{disease}} + 0.8 \mathcal{L}_{\text{severity}} + 0.2 \mathcal{L}_{\text{cons}}. \quad (6)$$

Training uses AdamW with an initial learning rate of $1e-4$, weight decay of 0.05, a three-epoch warm-up stage, and cosine learning-rate decay. The model is trained for 25 epochs on 224×224 images. Class-balanced sampling is applied for severity grading, while dropout, label smoothing, and early stopping are used to limit overfitting. The total parameter count of the final multi-task model is 29.63 million, which remains moderate for edge-oriented agricultural applications.

To reduce optimization conflict between disease and severity tasks, gradient magnitudes from the two heads are balanced during shared-backbone updates. This prevents one task from dominating the encoder and is especially important because the disease branch operates on 61 classes while the severity branch operates on only 3 classes. In practice, balanced updates stabilize convergence and make the joint model easier to tune.

The implementation also retains practical regularization strategies from the original solution, including exponential moving average for model parameters, gradient clipping, and conservative test-time augmentation. These settings are not conceptually complex, but they help produce stable validation curves and reduce sensitivity to noisy agricultural samples. For a conference-style paper, such details are useful because they show that the gains come from a reproducible pipeline.

E. Inference and Diagnostic Reporting

At inference time, the model produces both disease and severity posteriors in a single forward pass. The final disease label is obtained from the maximum disease probability, while the severity label is obtained from the three-class severity head. A simple consistency score can also be derived by reading the severity probability associated with the most likely disease class. This gives users a direct indication of whether the two branches agree.

The framework further supports qualitative interpretation through Grad-CAM visualizations generated from the disease and severity heads. Because the heads share the same encoder but optimize different objectives, their attention maps reveal both overlapping and task-specific evidence. In many samples, the disease branch concentrates on disease-type patterns such as lesion texture and margin morphology, while the severity branch focuses more strongly on the spatial spread and density of damaged regions.

This form of output is useful for deployment because farmers and technicians rarely trust a single class label without a visual explanation. A report containing disease identity, severity level, confidence values, and highlighted lesion regions is easier to interpret than isolated probabilities. The reporting mechanism, therefore, turns the model into a more practical decision-support component rather than a purely offline classification benchmark.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

Performance was evaluated using accuracy and macro F1-score for both tasks. Accuracy measures overall correctness, while macro F1-score gives equal importance to all classes and is therefore better suited to imbalanced agricultural datasets. The experiments used AdamW, an initial learning rate of $1e-4$, weight decay of 0.05, 25 training epochs, and a warm-up plus cosine-decay schedule.

A. Main Comparison Results

Table 2 compares the proposed multi-task framework with single-task baselines derived from the same dataset and feature extraction pipeline. The disease-only model achieves 86.74% accuracy and 0.8273 macro F1-score. After joint learning, disease recognition improves to 87.60% accuracy and 0.8564 macro F1-score. This performance gain is significant, as disease classification is the more challenging task in the benchmark, which involves 61 visually similar fine-grained categories.

TABLE II. COMPARISON BETWEEN SINGLE-TASK BASELINES AND THE PROPOSED MULTI-TASK MODEL

Method	Disease acc. (%)	Disease macro F1	Severity acc. (%)	Severity macro F1
Disease-only baseline	86.74	0.8273	-	-
Severity-only baseline	-	-	93.00	0.9200
Proposed multi-task model	87.60	0.8564	92.98	0.9338

On the severity task, the single-task baseline reaches 93.00% accuracy and 0.9200 macro F1-score, while the proposed multi-task model reaches 92.98% accuracy and a macro F1-score of 0.9338. Although absolute accuracy remains nearly unchanged, the higher macro F1-score indicates that the proposed model provides a more balanced treatment of the three severity classes. From a deployment viewpoint, this means that one shared model can replace two independent models without sacrificing grading quality.

B. Severity-level Analysis

TABLE III. CLASS-WISE F1-SCORE FOR SEVERITY GRADING IN THE FINAL MULTI-TASK MODEL

Item	Value
Healthy	0.9567
General	0.9138
Serious	0.9309
Macro average	0.9338

Table 3 reports F1-scores for each severity class in the final multi-task system. The Healthy class obtains the highest F1-

score of 0.9567, showing that the model can reliably separate healthy leaves from diseased ones. The General and Serious classes reach 0.9138 and 0.9309, respectively. The gap between these two classes is expected because moderate and advanced symptoms may share overlapping visual cues, especially when lesion spread is localized.

The strong performance on the Serious category is especially important from an application viewpoint. In agricultural management, underestimating a severe case is usually more costly than overestimating a mild one, because delayed intervention can accelerate yield loss and disease spread. The reported F1-score of 0.9309 suggests that the model remains sensitive to high-risk samples even though it is trained jointly with a much larger disease classification problem.

Qualitative attention analysis from the original experimental pipeline also supports the numerical results. Feature maps and Grad-CAM visualizations show that the network tends to focus on necrotic regions, discoloration boundaries, and high-contrast lesion areas rather than irrelevant backgrounds. This behavior strengthens the practical interpretability of the model.

C. Synergy and Diagnostic Consistency

Beyond standard accuracy metrics, the multi-task framework should also be judged by whether the two outputs agree in a meaningful way. The original experimental logs report an average cross-task consistency score of 0.9613, which indicates that the severity prediction is usually well aligned with the disease branch after the mapping from disease classes to severity levels is applied. This is precisely the behavior expected from the proposed consistency regularizer.

Another way to understand model behaviors is through interpretability statistics. For the disease head, the relevant lesion regions have an average attention coverage of 68.3%, and for the severity head, it is 72.1%. The two attention maps have an overlap ratio of 45.7%. This figure is sufficiently large to validate the existence of shared evidence, yet small to confirm the non-redundancy of the two tasks. The shared encoder extracts common pathology cues, but each head still captures task-specific features.

Along with a moderate parameter count, these deployment-oriented metrics indicate that the proposed method adds value beyond raw accuracy through coherent outputs, efficient shared representations, and intelligible lesion localization. All of these are critical for conference submissions focused on intelligent agriculture systems that are practical.

D. Practical Implications and Limitations

The proposed method offers three practical advantages. First, using a shared backbone means lower computational cost compared with separately deploying disease and severity models. Second, the consistency loss alleviates the occurrence of semantically nonsensical output combinations. Third, a model parameter scale of 29.63 million is moderately sized and keeps the model substantially lighter than most multi-stage agricultural pipelines that are built with independent segmentation, classification, and post-processing modules.

Despite these advantages, the study still has limitations. The dataset is image-centric and does not explicitly encode field context such as illumination and cultivar. Additionally, framing severity as a three-tier classification task, rather than as a continuous score, forfeits a degree of agronomic detail.

Future work can extend the present framework in several directions. Integrating lesion segmentation could provide clearer area-based grading. Uncertainty estimation could mark ambiguous samples for manual review. Model distillation could compress the shared backbone for mobile devices. Domain-adaptive pretraining on field imagery is particularly promising for closing the gap between laboratory-style images and open-environment captures [7].

V. CONCLUSION

This paper introduces a cross-task consistent multi-task framework for diagnosing and grading leaf diseases in crops at a fine-grained level. The model delivers coherent predictions in a single inference pipeline due to the integration of a ConvNeXt-Tiny encoder, CBAM attention refinement, parallel disease and severity heads, along with a lightweight consistency regularizer.

Experiments on a 61-class, 10-crop dataset show that the method improves disease recognition to 87.60% accuracy and 0.8564 macro F1-score, while maintaining strong severity grading performance at 92.98% accuracy and 0.9338 macro F1-score. These results suggest that multi-task learning is a practical way to build compact, accurate, and interpretable diagnosis systems for precision agriculture.

REFERENCES

- [1] M. Shafay, T. Hassan, M. Owais, et al., "Recent advances in plant disease detection: Challenges and opportunities," *Plant Methods*, vol. 21, art. no. 140, 2025.
- [2] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Front. Plant Sci.*, vol. 7, art. no. 1419, 2016.
- [3] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, New Orleans, LA, USA, 2022, pp. 11976–11986.
- [4] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Munich, Germany, 2018, pp. 3–19.
- [5] Z. Bi, F. Ma, J. Guan, et al., "Apple leaf disease severity grading based on deep learning and the DRL-Watershed algorithm," *Sci. Rep.*, vol. 15, art. no. 30071, 2025.
- [6] K. M. Sudhesh, R. Aarthi, P. S. Kurian, et al., "Interpretable multitask deep learning model for detecting and analyzing severity of rice bacterial leaf blight," *Sci. Rep.*, vol. 15, art. no. 27313, 2025.
- [7] R. Fu, X. Wang, S. Wang, and H. Sun, "PMJDM: A multi-task joint detection model for plant disease identification," *Front. Plant Sci.*, vol. 16, art. no. 1599671, 2025.
- [8] H. Zhu, D. Wang, Y. Wei, et al., "YOLOV8-CMS: A high-accuracy deep learning model for automated citrus leaf disease classification and grading," *Plant Methods*, vol. 21, art. no. 88, 2025.
- [9] X. Jiang, J. Wang, K. Xie, et al., "PlantCaFo: An efficient few-shot plant disease recognition method based on foundation models," *Plant Phenomics*, vol. 7, art. no. 100024, 2025.